ABRIDGED CHAIN OF THOUGHT FOR "IMPROVED LONG GAPS BETWEEN PRIMES"

I want to find a long interval containing no primes while keeping its starting point small. These are two separate demands. If I prescribe one residue class for each prime in a covering, the Chinese remainder theorem produces a starting point below the product of those primes [1, Lemma 1.1 and Section 3]. That is a convenient upper bound, but it may be a very poor measure of where the first successful interval actually occurs. The construction should be allowed to use divisors whose full product is much larger than the starting point I am trying to find.

Write

$$L = \log x, \qquad \ell = \log L, \qquad h = \log \ell.$$

The initial target is a composite interval of length

$$H = \left\lfloor \frac{AxL}{h} \right\rfloor$$

for every fixed $A > 0$, with starting point at most $e^{O(x)}$. After setting x to a small constant times $\log X$, this would give every fixed multiple of

$$\frac{\log X \ \log_2 X}{\log_4 X}$$

as a lower bound for the largest prime gap below X. It is useful to keep the dependence on the number of remaining offsets explicit: if the final argument can handle more of them, the preliminary interval can be made longer.

There is a reason to separate the covering modulus from the location. Suppose a preliminary sieve leaves $k = o(H)$ offsets in $[1, H]$, and consider primes $H < p \le H^2$. Each of these primes can hit at most one offset. Nevertheless, in the complete residue space, the probability that every offset gets a divisor is at least $e^{-O(k)}$. One can see this by assigning distinct primes to the targets and making every other prime miss all targets. The reciprocal mass of the available primes is bounded below, and excluding the primes already assigned only removes a small amount of that mass.

Thus there can be many complete covers even though the product of all available primes is enormous. If the successful residue classes were sufficiently well distributed, their first occurrence could be much earlier than that product. Abundance by itself does not prove this. A set occupying a substantial fraction of a period can still avoid a long initial interval. Related CRT equidistribution estimates average discrepancy over moduli [2, Theorems 1.4 and 1.6]. I need a certificate whose average over actual small starting points can be estimated, rather than only an estimate over a complete period.

The most direct certificate is a weight on starting points. If $P_H(N)$ counts the primes among $N+1, \dots, N+H$, then any nonnegative weight W satisfying

$$\sum_N W(N) P_H(N) < \sum_N W(N)$$

forces an empty interval: the prime count is a nonnegative integer, so it cannot be at least one everywhere that the weight is positive. A useful choice is the square of a signed arithmetic function. Cancellation can occur before squaring, while the final inequality still has a nonnegative weight.

This distinction matters for the error terms. Expanding a positive covering weight into separate congruence conditions and bounding every counting error in absolute value can cost the whole covering modulus again. A signed amplitude may have a much shorter useful expansion even when the function from which it is obtained involves a very large collection of primes. The problem becomes one of finding an amplitude concentrated on complete covers whose arithmetic complexity is small enough to transfer to a short interval.

A preliminary sieve can reduce the number of difficult offsets considerably. For the fixed-A target above, take

$$w = L^4, \qquad u_0 = \frac{5h}{\log h}, \qquad z = x^{1/u_0}.$$

Eventually $H/x < w < z < x$. Choose residue zero for primes $p \le w$ and for primes $z < p \le x$. For the intermediate primes $w < p \le z$, choose independent uniformly random nonzero residues.

It is important to count surviving composites as well as surviving primes. If a surviving composite $n \le H$ had a prime factor greater than x, its complementary factor would lie between 2 and $H/x < w$. That complementary factor would have a prime divisor removed by the small-prime zero sieve. A surviving integer cannot have a prime factor in $(z, x]$, either. Consequently every survivor is either a prime greater than x, or a z-smooth integer.

Put

$$\sigma = \prod_{w<p\le z} \frac{p-2}{p-1}.$$

A prime greater than x survives with probability σ. For a w-rough, z-smooth integer $n \le H$, the probability is

$$\sigma \prod_{p|n} \frac{p-1}{p-2}.$$

The product is over distinct prime divisors. Its logarithm is at most

$$O\left(\frac{\omega(n)}{w}\right) = O\left(\frac{\log H}{w \log w}\right) = o(1),$$

so this probability is $(1+o(1))\sigma$, uniformly in the candidate. Prime powers therefore cause no missing case.

The elementary estimates needed here are

$$ct \le \vartheta(t) \le 4t, \qquad \pi(t) \asymp \frac{t}{\log t},$$

$$\sum_{p\le t} \frac{\log p}{p} = \log t + O(1), \qquad \sum_{p \le t} \frac{1}{p} = \log\log t + b + O(1/\log t).$$

The Chebyshev bounds follow from the central binomial coefficient, and the first prime sum follows by expanding $\log(n!)$ in prime powers. The rounding error is $O(\psi(n)) = O(n)$, and the contribution of powers of exponent at least two is also $O(n)$. Partial summation then gives the reciprocal-prime estimate. In particular,

$$\prod_{a<p\le b} (1-1/p)^{-1} \asymp \frac{\log b}{\log a}, \qquad \sum_{p>a} \frac{1}{p^2} \ll \frac{1}{a \log a}.$$

These estimates give $\sigma \ll u_0 \ell / L$.

For the smooth numbers, let $u = \log H/\log z$ and put $\beta = \log u/\log z$. If n_z denotes the full z-smooth part of n, then

$$n_z^\beta = \sum_{d|n} g(d), \qquad g(p^a) = p^{a\beta} - p^{(a-1)\beta} \quad (p \le z),$$

where g is nonnegative and multiplicative, and is zero on powers of primes above z. Therefore

$$\sum_{n\le M} n_z^\beta \le M \prod_{p\le z} \frac{1-1/p}{1-p^{\beta-1}} \le M e^{O(u)}.$$

Expanding the logarithm of the Euler product and using the prime sums gives the last bound. On a dyadic interval, a smooth integer has $n_z = n$, so the moment bound yields

$$\Psi(H,z) \leq 4H\exp(-u\log u + O(u)).$$

Here $u = (1+o(1))u_0$, and $u\log u = (5+o(1))h$. Hence

$$\Psi(H,z) \leq H\ell^{-5+o(1)}.$$

The expected survivor count is at most

$$\begin{aligned}\sigma\big(\pi(H) + (1+o(1))\Psi(H,z)\big) &= O_A\left(\frac{x\ell}{L\log h}\right) + O_A\left(\frac{x\ell^{-4+o(1)}}{\log h}\right)\\ &= o(x/\ell^2).\end{aligned}$$

Thus some deterministic choice leaves

$$k \leq x/\ell^2$$

uncovered offsets. The sieve is only a reduction to a smaller set; there is no need to assign a separate cleanup prime to each remaining integer.

Let

$$Q_0 = \prod_{p\leq x} p.$$

The chosen residue classes give $0 \leq b < Q_0$ such that every starting point

$$N_t = b + Q_0 t$$

already makes all but a set

$$S = \{s_1,\ldots,s_k\} \subseteq [1,H]$$

divisible by primes at most x. I want to finish the construction for some $1 \leq t \leq e^x$. The unresolved task is to make the remaining $N_t + s_i$ composite without paying for the product of all primes used to do it.

Take

$$y = \exp(x/L^2), \qquad \mathcal{P} = \{p : H < p \leq y\}.$$

Eventually $H^2 < y < Q_0$. For any fixed $p \in \mathcal{P}$, the roots in t corresponding to the offsets s_i are distinct, because $p > H$ and $p \nmid Q_0$. In a complete CRT period, the state of this source prime is consequently

$$\Pr(X_p = i) = 1/p \quad (1 \leq i \leq k), \qquad \Pr(X_p = 0) = 1 - k/p.$$

The source primes are independent. The targets are not independent: at a given prime, hitting one excludes hitting the others [3, Section 1.3 and proof of Lemma 6.1]. This categorical law is the model whose estimates must eventually be transferred to actual integers.

For one target, its hit indicators I_p are independent Bernoulli variables of means $1/p$. The natural orthogonal functions are

$$\psi_p = \frac{1-pI_p}{p-1}, \qquad \psi_d = \prod_{p\mid d}\psi_p,$$

where throughout these expressions d is squarefree and supported on $\mathcal{P}$. They satisfy

$$\mathbb{E}\psi_d\psi_e = 0 \quad (d \neq e), \qquad \mathbb{E}\psi_d^2 = \frac{1}{\varphi(d)}.$$

The empty hit set is a distinguished boundary point. At that point,

$$\psi_d(\varnothing) = \frac{1}{\varphi(d)}.$$

I would like a function with mean one which vanishes there and has small energy when a source prime is charged $\log p$.

For a one-target expansion $f = 1 + \sum_{d>1} a_d \psi_d$, the vanishing condition and energy are

$$1 + \sum_{d>1} \frac{a_d}{\varphi(d)} = 0, \qquad \mathscr{E}(f) = \sum_{d>1} \frac{\log d}{\varphi(d)} a_d^2.$$

Weighted Cauchy–Schwarz suggests using every divisor scale with coefficient proportional to $1/\log d$. Set

$$c_* = \sum_{d>1} \frac{1}{\varphi(d)\log d}, \qquad f = 1 - \frac{1}{c_*} \sum_{d>1} \frac{\psi_d}{\log d}.$$

Then

$$f(\varnothing) = 0, \qquad \mathbb{E} f = 1, \qquad \mathscr{E}(f) = \frac{1}{c_*}.$$

Indeed, for any mean-one function with that boundary value, Cauchy–Schwarz gives $1 \le c_* \mathscr{E}(f)$, and this choice attains equality. The reciprocal logarithm is therefore forced by the energy minimization, rather than being an arbitrary smooth weight.

The size of c_* is the decisive point. Using only individual prime hits gives an energy cost of order $\log H$. Summing across divisor scales can reduce that cost by a factor comparable to the logarithmic range of the sources. To see the gain, write

$$J(t) = \prod_{p\in\mathcal{P}} \left(1 + \frac{p^{-t}}{p-1}\right).$$

The identity $1/\log d = \int_0^\infty d^{-t}\,dt$ gives

$$c_* = \int_0^\infty (J(t) - 1)\,dt.$$

For

$$1/\log y \le t \le 1/\log H,$$

split the product at $e^{1/t}$. Below that point, replacing p^{-t} by one changes the logarithm of the product by $O(1)$. Above it, the remaining logarithmic contribution is bounded. Consequently

$$J(t) \asymp \frac{1}{t \log H}$$

on this range. Its integral contributes

$$\frac{\log(\log y/\log H)}{\log H}.$$

The interval $0 < t < 1/\log y$ contributes $O(1/\log H)$, since $J(0) \asymp \log y/\log H$. For $t > 1/\log H$, the prime sum bounds give $J(t) - 1 \ll e^{-t\log H}$, so that range contributes the same order. Thus

$$c_* \asymp \frac{\log(\log y/\log H)}{\log H} \asymp 1.$$

Here $\log H \asymp L$ while $\log(\log y/\log H) = L - 3\ell + O(1)$. The energy of one target is bounded by an absolute constant, even though the primes range over many scales.

Low energy alone is not enough. I also need positivity, a uniform upper bound, and some control of the function on nonempty hit sets. These are clearer in an integral representation than in the signed orthogonal expansion. Define

$$r_p(t) = \frac{1 - p^{-t}}{1 + p^{-t}/(p-1)}.$$

If E is the set of primes hitting the target, then the corresponding Euler product equals $J(t)\prod_{p\in E} r_p(t)$. Subtracting it from the empty-hit product gives

$$f(E) = \frac{1}{c_*}\int_0^\infty J(t)\left(1 - \prod_{p\in E} r_p(t)\right)\, dt.$$

Every $r_p(t)$ lies in $[0,1]$. Hence f is nonnegative, increases as hits are added, and vanishes when there are no hits. Its increments decrease as the existing hit set grows. This last fact will make the categorical dependence manageable.

The maximum occurs when all primes hit. In that case the second product in the difference is $\prod_p(1-p^{-t})$, so

$$f(\mathcal{P}) = 1 + \frac{1}{c_*}\int_0^\infty \left(1 - \prod_{p\in\mathcal{P}}(1-p^{-t})\right)\, dt.$$

Bound the integral over $[0,1]$ by one, and use the sum of the p^{-t} terms after that. This gives

$$f(\mathcal{P}) \le 1 + \frac{1}{c_*}\left(1 + \sum_{p\in\mathcal{P}} \frac{1}{p\log p}\right) = O(1).$$

Thus $0 \le f \le M$ for an absolute constant M.

For a nonempty E, retain a single hit in the integral and restrict to $0 < t < 1/\log y$. In this range $J(t) \gg J(0)$ and $1 - r_p(t) \ge e^{-1}$. Therefore

$$f(E) \gg \frac{J(0)}{c_*\log y} \gg \frac{1}{\log H} \gg \frac{1}{L}.$$

The useful collection of facts is now

$$f(\varnothing) = 0, \quad \mathbb{E}f = 1, \quad 0 \le f \le M, \quad f(E) \ge c/L\ (E \ne \varnothing), \quad \mathscr{E}(f) = 1/c_* = O(1).$$

The function is spread over the available prime scales and has a cheap boundary condition. Multiplying such functions is a plausible way to impose all the target conditions at once.

In the categorical source model, let

$$F = \prod_{i=1}^k f_i, \qquad Z = \mathbb{E}F^2.$$

The function F vanishes whenever any target is unhit. If the targets were independent, the product energy would be easy to estimate. They are not, and replacing the source law by independent target columns would lose exactly the dependence that comes from actual congruences. I need to estimate Z and the energy in the true law.

First, a fairly weak lower bound for Z is enough to control the effect of tilting the probability measure. Use just the source primes in $(H, H^2]$. Their reciprocal sum tends to $\log 2$. The probability of no hits in this band is

$$\prod_{H<p\le H^2}(1-k/p) = e^{-O(k)},$$

since $k = o(H)$. Starting from that event, assign one distinct prime to each target. At a chosen prime, the likelihood ratio for a prescribed hit in place of a null state is $1/(p-k)$. The sum of these weights remains bounded below after excluding fewer than k already chosen primes: the excluded mass is $O(k/H) = o(1)$. Summing over distinct assignments gives a complete-cover probability at least $e^{-O(k)}$.

On this event every factor f_i is at least c/L. Thus

$$Z \ge e^{-O(k)}(c/L)^{2k} = e^{-O(k\ell)}.$$

The same argument works with any $r \le k$ targets and with one source prime deleted. Deleting one prime changes the available reciprocal mass negligibly. The function f is kept fixed when a source is deleted; that source is simply forced to be absent from the hit set. This uniformity is needed when estimating individual source gradients.

Write

$$F_r = \prod_{i=1}^{r} f_i, \qquad Z_r = \mathbb{E} F_r^2,$$

and consider the r-target law tilted by F_r^2/Z_r. The quantity that measures interference with another target is

$$A_{\text{used}} = \sum_{\substack{p:\ p \text{ hits an} \\ \text{existing target}}} \frac{1}{p}.$$

It is the reciprocal mass of the sources made unavailable by existing hits. Bounding the number of occupied sources would be unnecessarily expensive; their reciprocal mass is what changes the new target's distribution.

Under the untilted law, the events of a source being used are independent and have probabilities r/p. Since $p > H$,

$$\begin{aligned} \log \mathbb{E} e^{H A_{\text{used}}} &\le \sum_{p \in \mathcal{P}} \frac{r}{p}(e^{H/p} - 1) \\ &\le (e-1) r H \sum_{p > H} \frac{1}{p^2} \ll \frac{r}{\log H}. \end{aligned}$$

Meanwhile $F_r \le M^r$ and the cover lower bound give a tilted density at most $e^{O(r\ell)}$. Combining the density bound with the exponential moment gives

$$\Pr_{\text{tilt}}(A_{\text{used}} > a) \le \min\{1, \exp(O(r\ell) - Ha)\}.$$

Integrating this tail shows, uniformly in $r \le k$ and in the deletion of one source,

$$\mathbb{E}_{\text{tilt}} A_{\text{used}} = o(1), \qquad \mathbb{E}_{\text{tilt}} e^{2A_{\text{used}}} \le 2,$$

provided $k\ell/H \to 0$. For $H = \lfloor AxL/h \rfloor$ and $k \le x/\ell^2$, this condition has considerable room to spare.

Now insert target r, initially collapsed into the null category of the $(r-1)$-target model. Conditional on the existing target profiles, each unused prime hits the new target independently with probability

$$\frac{1}{p - r + 1} \ge \frac{1}{p}.$$

A used prime cannot hit it. Compared with the one-target Bernoulli law, increasing probabilities at unused primes helps because f is increasing. Removing the used primes costs at most MA_{used} in the mean, since a change at any one coordinate changes f by at most M. Thus the conditional mean of the new factor is at least

$$1 - MA_{\text{used}}.$$

If a source p_0 has also been deleted, allow the additional loss M/p_0.

Average under the tilted old law and use Jensen's inequality. The conditional second moment of the new factor dominates the square of its conditional first moment, and the averaged first moment is $1 - o(1)$. It follows that

$$\frac{Z_r}{Z_{r-1}} \ge 1 - o(1) \ge \frac{1}{2}.$$

The deleted-source version holds as well. This improves the crude cover lower bound to

$$Z \ge 2^{-k}.$$

The crude bound was useful for starting the comparison; the ratios are stronger and are what keep the later estimates from accumulating a large cost with every target.

There is a second comparison that will be needed when one target is singled out. Condition on target i being unhit. At each prime, each of the other targets then has probability $1/(p-1)$. Relative to the ordinary $(k-1)$-target law, a used prime has likelihood ratio

$$\frac{p}{p-1},$$

while the likelihood ratio at a null prime is at most one. The total density ratio is therefore at most $e^{2A_{\mathrm{used}}}$. If G_0 is the product of the other detector factors under this conditional law, the exponential-moment estimate and the partition ratio give

$$\mathbb{E}G_0^2 \le 2Z_{k-1} \le 4Z.$$

Conditioning a target to be unhit is a rare event, but the other factors have a second moment comparable to Z under that condition. Dividing by the rarity of the event would give a much worse estimate.

The product energy can now be controlled. I use the source-coordinate energy

$$\mathscr{E}(F) = \sum_{p\in\mathcal{P}} \log p\, \mathbb{E}\operatorname{Var}_p(F),$$

where Var_p resamples only the categorical state at p. Fix the other sources. If F_0 is the value with p null and F_j the value when it hits target j, then

$$\operatorname{Var}_p(F) \le \frac{1}{p}\sum_{j=1}^{k}(F_j - F_0)^2.$$

Changing null to a hit at j changes only its detector factor, so

$$F_j - F_0 = (\Delta_p f_j)\prod_{i\ne j} f_i.$$

The relevant question is how the other target profiles affect the squared increment $(\Delta_p f_j)^2$.

The integral representation gives

$$\Delta_p f(E) = \frac{1}{c_*}\int_0^\infty J(t)(1-r_p(t))\prod_{q\in E} r_q(t)\,dt,$$

with p omitted from the old hit set. Its square is a positive double integral whose dependence on each old hit at q is through

$$r_q(t)r_q(s)\in[0,1].$$

It is therefore decreasing in the other hit indicators. Conditional on the old profiles, the increased hit probabilities at unused primes reduce its expectation. At a used prime q, removing the Bernoulli hit can increase the expectation by at most

$$(1-1/q)^{-1}.$$

The product of those losses is at most $e^{2A_{\mathrm{used}}}$, whose tilted mean is bounded.

Let m_p be the ordinary one-target expectation of $(\Delta_p f)^2$, with source p omitted. The preceding comparison bounds the average squared product gradient by a constant times m_p times the appropriate $(k-1)$-target normalizer. The insertion ratios, including the deleted-source ratios, compare that normalizer to Z. Keeping a convenient constant gives

$$\frac{\mathscr{E}(F)}{Z} \le 8k\sum_{p\in\mathcal{P}} \frac{\log p}{p} m_p.$$

For the one-target law,

$$\mathscr{E}(f) = \sum_p \frac{1}{p}\left(1 - \frac{1}{p}\right) \log p \, m_p = \frac{1}{c_*}.$$

Since $p > H$, the factor $(1 - 1/p)^{-1}$ is at most two. Hence

$$\mathscr{E}(F) \le \frac{16k}{c_*} Z \ll kZ.$$

The energy cost is linear in the number of targets, with an absolute constant. The categorical exclusions have been accounted for through the mass of used primes, without pretending that the target columns are independent.

The full product F has the desired zeros but depends on an enormous period. I need to cut down its source products. Decompose it orthogonally by sets of source coordinates:

$$F = \sum_{\mathcal{A}} F_{\mathcal{A}}, \qquad w(\mathcal{A}) = \sum_{p \in \mathcal{A}} \log p.$$

Each $F_{\mathcal{A}}$ has mean zero in every coordinate in $\mathcal{A}$, and is constant in all other coordinates. Then

$$Z = \sum_{\mathcal{A}} \|F_{\mathcal{A}}\|_2^2, \qquad \mathscr{E}(F) = \sum_{\mathcal{A}} w(\mathcal{A}) \|F_{\mathcal{A}}\|_2^2.$$

Large source products are expensive in energy. A cutoff at logarithmic cost of order x should retain much of the norm because the average cost is only $O(k)$.

A sharp projection is not the most useful cutoff. From its ordinary squared error one would only get a bound for the uncovered count by multiplying by k, which is too costly. What I need is control of the filtered function on each boundary where a target is unhit, and then a way to sum those boundary estimates. A cutoff that varies gradually with source cost gives the required difference bound.

Set

$$B = x/100, \qquad D = e^B, \qquad \theta(s) = (1 - s/B)_+,$$

and define

$$R = \sum_{\mathcal{A}} \theta(w(\mathcal{A})) F_{\mathcal{A}}.$$

Every source product appearing in R is at most D. The function R can have either sign; the final weight will be R^2. Orthogonality and

$$1 - (1 - s/B)_+^2 \le 2s/B$$

immediately give

$$\mathbb{E}R^2 \ge Z - \frac{2}{B}\mathscr{E}(F).$$

For $k \le x/\ell^2$, this is $(1 - o(1))Z$.

Let U_i indicate that target i is unhit, and let $U = \sum_i U_i$. The important bound is

$$\mathbb{E}[UR^2] \le \frac{\gamma J_1}{B^2}\mathscr{E}(F), \qquad J_1 = \sum_{p \in \mathcal{P}} \frac{\log p}{p}, \qquad \gamma = \max_{p \in \mathcal{P}} \frac{p}{p-1}.$$

The factor B^{-2} comes from the slope of the cutoff. The sum J_1, rather than a multiple of k from a pointwise bound on U, is what makes this estimate useful.

To understand it, fix a target i. At source p, choose an orthonormal basis consisting of the constant function, the vector

$$v_{i,p} = \frac{1 - p\,1_{\{X_p = i\}}}{\sqrt{p-1}},$$

and complementary mean-zero vectors. Each complementary vector vanishes at the state $X_p = i$: orthogonality to the constant and to $v_{i,p}$ forces this. Group a product-basis expansion by the set

J of sources at which complementary vectors occur, and by the choices of those vectors. On the remaining sources a block has a binary expansion

$$\sum_d a_d v_d.$$

The complementary part uses cost $c = \sum_{p\in J} \log p$, while the binary factor indexed by d uses cost $\log d$.

Since F is zero whenever target i is unhit, the binary coefficients in each such block satisfy

$$\sum_d \frac{a_d}{\sqrt{\varphi(d)}} = 0.$$

This is the original vanishing condition expressed in the basis adapted to that target. After filtering, the block's value on the same boundary is

$$\sum_{d>1} a_d \frac{\theta(c+\log d) - \theta(c)}{\sqrt{\varphi(d)}}.$$

Subtracting $\theta(c)$ is legitimate precisely because the unfiltered value was zero. It turns the estimate into one about a difference of cutoff values.

Cauchy–Schwarz with energy weights gives the square at most

$$\left(\sum_{d>1} \log d\, |a_d|^2\right)\left(\sum_{d>1} \frac{|\theta(c+\log d) - \theta(c)|^2}{\varphi(d)\log d}\right).$$

The Lipschitz bound $|\theta(s) - \theta(t)| \le |s-t|/B$ bounds this by

$$\frac{1}{B^2}\left(\sum_{d>1} \log d\, |a_d|^2\right)\left(\sum_{d>1} \frac{\log d}{\varphi(d)}\right).$$

All divisors in this block use sources outside J. Multiplying the second factor by the probability that these sources all miss target i simplifies it exactly:

$$\left(\prod_{p\notin J}(1-1/p)\right) \sum_{\substack{d>1\\ p|d\Rightarrow p\notin J}} \frac{\log d}{\varphi(d)} = \sum_{p\notin J} \frac{\log p}{p} \le J_1.$$

The large Euler product cancels against the unhit probability. Distinct complementary blocks remain orthogonal on the unhit event, so their estimates can be added.

There is still the sum over targets. At a fixed source p, the Gram matrix of the vectors $v_{i,p}$ has diagonal entries one and off-diagonal entries

$$\langle v_{i,p}, v_{j,p}\rangle = -\frac{1}{p-1} \qquad (i \ne j).$$

Its largest eigenvalue is at most $p/(p-1)$. Therefore the sum of the binary-direction energies over all targets is bounded by γ times the full source energy. This proves the displayed estimate for $\mathbb{E}[UR^2]$. The negative correlations do not disappear; they are encoded in a Gram matrix with a uniformly bounded norm.

Since

$$J_1 \ll \log y = x/L^2,$$

the energy estimate yields

$$\mathbb{E}R^2 = (1-o(1))Z, \qquad \mathbb{E}[UR^2] \ll \frac{Z}{\ell^2 L^2} = o(Z).$$

The filtered amplitude has retained substantial norm and gives very little weight to unhit targets in the complete residue model. This is the desired geometric effect, but it is not yet an arithmetic estimate over $1 \le t \le e^x$.

The unhit indicator involves every source prime, so multiplying R^2 by it brings back the very large period. I need a finite divisor expression that equals one at an unhit target and whose square can majorize a prime indicator. For each target define

$$H_D = \sum_{d \le D} \frac{1}{\varphi(d)}, \qquad S_i = \frac{1}{H_D} \sum_{d \le D} \psi_{i,d}.$$

All divisors remain squarefree and supported on $\mathcal{P}$. If the target is unhit, each $\psi_{i,d} = 1/\varphi(d)$, so $S_i = 1$. In particular, at an actual prime greater than y, the square S_i^2 is one [4, Appendices 7–8]. The finite score to transfer is consequently $\sum_i S_i^2$.

Put $e_i = S_i - U_i$. It would suffice to prove

$$\sum_i \mathbb{E}[R^2 e_i^2] = o(Z),$$

because

$$S_i^2 \le 2U_i + 2e_i^2.$$

An unweighted error estimate for $S_i - U_i$ is not enough by itself: the weight R^2 might concentrate on the exceptional set. Using the global bound $F \le M^k$ at this step would introduce an exponential factor in k. The error must be estimated relative to Z by isolating one target at a time.

First consider the error in the one-target law. Let

$$M_0 = \prod_{p \in \mathcal{P}} \frac{p}{p-1}, \qquad u = \frac{\log D}{\log y} = \frac{L^2}{100}.$$

The normalized divisor weights $1/(M_0\varphi(d))$ form a probability distribution: independently at each prime, include p in d with probability $1/p$. The missing mass beyond D is

$$\tau = 1 - \frac{H_D}{M_0}.$$

With $\beta = \log u / \log y$, Rankin's inequality gives

$$\tau \le D^{-\beta} \mathbb{E} d^\beta.$$

For the divisor distribution,

$$\log \mathbb{E} d^\beta \le \sum_{p \in \mathcal{P}} \frac{p^\beta - 1}{p} = O(u).$$

This last estimate follows by expanding the exponential and using the logarithmically weighted prime sums; the restriction to primes above H only decreases the sum. Since $\beta \log D = u \log u$,

$$\tau \le \exp(-u \log u + O(u)).$$

Choosing $y = e^{x/L^2}$ leaves c_* bounded below while making u as large as a constant times L^2. The discarded divisor mass is consequently much smaller than any fixed power of x^{-1}. This choice of source cutoff is useful for the arithmetic score as well as for the energy.

There is an exact identity behind the approximation to U_i. For one target,

$$\sum_d \psi_d = \prod_{p \in \mathcal{P}} (1 + \psi_p) = M_0 U_i.$$

If a prime hits, its factor is zero; if no prime hits, the product is M_0. Write

$$K_{\le} = \sum_{d \le D} \psi_d, \qquad K_{>} = \sum_{d > D} \psi_d.$$

Then

$$e_i = M_0^{-1}\left(\frac{\tau}{1-\tau}K_{\le} - K_{>}\right).$$

The two errors are explicit: the small renormalization from replacing M_0 by H_D, and the tail of the full expansion. A fourth-moment estimate for these expressions will be strong enough to accommodate the weight.

At a single prime,

$$\mathbb{E}\psi_p = 0, \qquad |\mathbb{E}\psi_p^r| \le \frac{1}{p-1} \quad (2 \le r \le 4).$$

In an expansion of a fourth power, a prime that occurs in only one of the four divisor positions therefore contributes zero. There are

$$\binom{4}{2} + \binom{4}{3} + \binom{4}{4} = 11$$

nonempty incidence patterns of size at least two. Taking absolute values of the remaining local moments and dropping the divisor cutoffs gives

$$\mathbb{E}K_{\le}^4 \le \prod_p \left(1 + \frac{11}{p-1}\right) \le M_0^{11}.$$

The large number of divisors has turned into a fixed Euler-product exponent.

For $K_>$, every divisor in a fourth-moment term exceeds D. It is enough to retain that condition for the first divisor and use

$$1_{\{d_1 > D\}} \le (d_1/D)^{\beta}.$$

Of the eleven non-singleton incidence patterns, seven contain the first position and four do not. Therefore

$$\mathbb{E}K_>^4 \le D^{-\beta} \prod_p \left(1 + \frac{4 + 7p^{\beta}}{p-1}\right).$$

The excess over the previous Euler product is bounded by

$$\exp\left(O\left(\sum_p \frac{p^{\beta}-1}{p-1}\right)\right) = e^{O(u)}.$$

It follows that

$$\mathbb{E}K_>^4 \le M_0^{11} \exp(-u\log u + O(u)).$$

Combining this with the formula for e_i and the bound for τ gives

$$\mathbb{E}e_i^4 \ll M_0^7 \exp(-u\log u + O(u)).$$

The factor M_0^{-4} comes from the normalization in e_i. The particular exponent seven is not the source of the gain; what matters is that it is absolute while the negative exponential has size $L^2 \log L$.

The remaining issue is how to bring R^2 into this estimate. Applying Cauchy–Schwarz directly with the scalar fourth moment of the full product would mix in all k target factors. Instead, fix i and expose its hit indicators through an enlarged source space.

For each prime, take independent variables ξ_p and Y_p, where

$$\Pr(\xi_p = 1) = 1/p,$$

and Y_p takes each other target label with probability $1/(p-1)$, and the null label with probability $(p-k)/(p-1)$. Set

$$X_p = i \quad \text{if } \xi_p = 1, \qquad X_p = Y_p \quad \text{if } \xi_p = 0.$$

This reproduces the categorical law exactly. The variables ξ_p give the ordinary independent one-target law, and the remaining labels are carried by Y. In this representation, e_i depends only on ξ.

Let $G_0(Y)$ be the product of the other detector factors when every $\xi_p = 0$. Turning some ξ_p to one removes hits from other targets and gives them to target i. By monotonicity of the detectors and $f_i \le M$,

$$0 \le F(\xi, Y) \le M G_0(Y).$$

The law of Y is exactly the law of the other targets conditional on target i being unhit. The anchored comparison already proved says

$$\mathbb{E}_Y G_0^2 \le 4Z.$$

Thus the bound for one isolated target costs only a constant times $\sqrt{Z}$ in $L^2(Y)$, instead of the global factor M^k.

Expand in the ξ-basis,

$$F(\xi, Y) = \sum_d b_d(Y)\psi_d(\xi).$$

Orthogonality gives

$$b_d(Y) = \varphi(d)\mathbb{E}_\xi[F(\xi, Y)\psi_d(\xi)].$$

For a prime, $\mathbb{E}|\psi_p| = 2/p$, so

$$\varphi(d)\mathbb{E}|\psi_d| \le 2^{\omega(d)}.$$

Using the pointwise bound by MG_0, followed by its second-moment bound, gives

$$\|b_d\|_{L^2(Y)} \le 2M\sqrt{Z}\, 2^{\omega(d)}.$$

This controls every coefficient by a factor depending on the divisor's number of prime factors, with no exponential loss in the number of targets.

It remains to check that filtering does not destroy this coefficient bound. The filter was defined by source coordinates, so in the enlarged space a source is nonconstant when either its ξ-coordinate or its Y-coordinate is nonconstant. Expand Y by source sets as well. If the ξ-frequency is d and the Y-frequency is e, the source cost is

$$\log \operatorname{lcm}(d, e).$$

A prime appearing in both coordinates is counted once. For a fixed d, the filter is diagonal on the orthogonal Y-components, with multipliers

$$\theta(\log \operatorname{lcm}(d, e)) \in [0, 1].$$

It therefore contracts the $L^2(Y)$-norm of each coefficient. The same coefficient bound holds for R.

This is why the source decomposition is important. Applying a target-by-target filter would not respect the shared-prime structure, and charging the two enlarged coordinates separately would use the wrong cost. The original source filter acts consistently after the enlargement because averaging an entire source in either representation is the same operation.

Consider now the fourth moment of the $L^2(Y)$-norm of R, viewed as a function of ξ. On expanding it, each term has two inner products of coefficient vectors. Their absolute values are bounded by the product of four coefficient norms. Singleton prime incidences still vanish under the ξ-expectation. A prime occurring in r of the four positions contributes at most $2^r/(p-1)$, so the local combinatorial factor is

$$\sum_{r=2}^{4} \binom{4}{r} 2^r = 72.$$

Consequently

$$\mathbb{E}_\xi \|R(\xi, \cdot)\|_{L^2(Y)}^4 \le 16M^4Z^2 \prod_p \left(1 + \frac{72}{p-1}\right) \le 16M^4Z^2M_0^{72}.$$

The estimate is deliberately generous. Its strength is that the normalizer appears as Z^2 and the rest is only a fixed power of M_0.

Because e_i is a function of ξ alone,

$$\mathbb{E}[R^2 e_i^2] = \mathbb{E}_\xi \left[e_i^2 \|R(\xi,\cdot)\|_{L^2(Y)}^2 \right]$$
$$\leq (\mathbb{E}_\xi e_i^4)^{1/2} \left(\mathbb{E}_\xi \|R(\xi,\cdot)\|_{L^2(Y)}^4 \right)^{1/2}.$$

The two fourth-moment estimates give

$$\frac{\mathbb{E}[R^2 e_i^2]}{Z} \ll M_0^{79/2} \exp\left(-\frac{1}{2} u \log u + O(u) \right).$$

Here

$$M_0 \asymp \frac{\log y}{\log H}, \qquad \log M_0 = O(L), \qquad u = L^2/100.$$

The logarithm of the right-hand side is at most $O(L) - cL^2 \log L + O(L^2)$. It tends to minus infinity fast enough to sum over every $k \leq x$, since $\log k \leq L$. Therefore

$$\sum_i \mathbb{E}[R^2 e_i^2] = o(Z).$$

The finite sieve approximation is now valid under the actual global weight, with the dependence on the other targets fully included.

Combining it with the boundary estimate gives

$$\mathbb{E}\left[R^2 \sum_i S_i^2 \right] = o(Z), \qquad \mathbb{E}R^2 = (1 - o(1))Z$$

for the initial survivor budget. This is the desired inequality in the complete categorical model. Every part of the score now has a finite expansion with source products at most D, which is what makes an arithmetic transfer possible.

Take

$$T = \lfloor e^x \rfloor$$

and evaluate the functions at the actual residues of $N_t + s_i$, for $1 \leq t \leq T$. A product of literal residue-class indicators in t is either incompatible or is a single class modulo the least common multiple of the source moduli. Its average over $[1, T]$ differs from its complete-period density by at most $1/T$. This estimate has no dependence on the modulus, but it must be multiplied by the absolute coefficient mass of the expansion. Controlling that mass is the reason for filtering the amplitude.

For a source set $\mathcal{A}$, its orthogonal component can be written by inclusion and exclusion as a signed sum of $2^{|\mathcal{A}|}$ conditional expectations of F. Each such conditional expectation lies between zero and M^k. If the source product is d, any function on those residues can be expanded into at most d literal residue-class indicators. Thus one component with source product d has coefficient mass at most

$$M^k d \, 2^{\omega(d)}.$$

Only $d \leq D$ are retained in R, and there are at most D possible integer values of d.

Since every source prime exceeds H,

$$\omega(d) \leq \frac{\log D}{\log H}, \qquad 2^{\omega(d)} \leq D^{\log 2/\log H} = D^{o(1)}.$$

The absolute coefficient sum for R therefore satisfies

$$L_R \leq M^k D^{2+o(1)}.$$

The bound uses the number of residue classes in a source product, rather than the number of possible target labelings across the full source pool. No factor proportional to the whole period is present.

For the sieve amplitude, expand

$$\psi_{i,d} = \prod_{p|d} \frac{1 - pI_{i,p}}{p-1}$$

in literal divisibility indicators. Its coefficient mass is

$$\prod_{p|d} \frac{p+1}{p-1} = D^{o(1)}.$$

There are at most D divisors in the sum for S_i, and $H_D \geq 1$, so

$$L_S \leq D^{1+o(1)}.$$

These bounds are coarse but sufficient because $D = e^{x/100}$ is a small fixed power of the sampling range e^x.

Coefficient mass is submultiplicative. Applying the $1/T$ error to the expansions of R^2 and $R^2 \sum_i S_i^2$ gives

$$\left| \mathrm{Avg}_{t \leq T} R(t)^2 - \mathbb{E}R^2 \right| \leq \frac{L_R^2}{T},$$

$$\left| \mathrm{Avg}_{t \leq T} \left(R(t)^2 \sum_i S_i(t)^2 \right) - \mathbb{E} \left(R^2 \sum_i S_i^2 \right) \right| \leq \frac{k L_R^2 L_S^2}{T}.$$

For $k \leq x/\ell^2$, the larger error is

$$e^{-.94x + o(x)}.$$

Even the crude lower bound $Z \geq e^{-O(k\ell)}$ is $e^{-o(x)}$, so both errors are $o(Z)$. Thus the strict score inequality survives on actual small values of t:

$$\mathrm{Avg}_{t \leq T} \left(R(t)^2 \sum_i S_i(t)^2 \right) < \mathrm{Avg}_{t \leq T} R(t)^2,$$

and the right-hand side is positive.

Some t with $R(t)^2 > 0$ must therefore satisfy $\sum_i S_i(t)^2 < 1$. A prime value $N_t + s_i$ is larger than y, because $N_t \geq Q_0 > y$. It has no divisor in $\mathcal{P}$, which would make $S_i(t) = 1$, contradicting that inequality. Hence every remaining target is composite. In fact, the same reasoning shows that each remaining target is hit by a source prime: any unhit target, prime or composite, would also have $S_i = 1$.

All offsets outside S already have divisors at most x, and these divisors are proper since the translated integers are much larger than x. The source divisors are proper as well, since they are at most $y < N_t$. Consequently

$$N_t + 1, \ldots, N_t + H$$

are all composite. If $k = 0$, the preliminary sieve has already achieved this and $t = 1$ suffices.

The height estimate only uses the preliminary modulus and the range of t:

$$N_t \leq Q_0(T+1) \leq 2e^{5x},$$

using $\vartheta(x) \leq 4x$. The product of all primes in $\mathcal{P}$ never enters this height bound. Their full residue space was used to prove an inequality, and the short expansion of the filtered amplitude was used to transfer it to the desired range of starting points.

For every fixed $A > 0$, this gives a composite block of length $\lfloor AxL/h \rfloor$ below $2e^{5x}$. To obtain a statement about prime gaps with both endpoints below X, set

$$x = \frac{\log X}{6}, \qquad A = 12C.$$

Then $N_t \le 2X^{5/6}$, and

$$\frac{H}{\log X \, \log_2 X / \log_4 X} \longrightarrow \frac{A}{6} = 2C.$$

Let p be the largest prime at most N_t, and q the smallest prime greater than $N_t + H$. They are consecutive, and $q - p \ge H + 1$.

The elementary Chebyshev bounds imply that for some fixed sufficiently large K, every sufficiently large interval $(v, Kv]$ contains a prime: choose K so that the lower bound for $\vartheta(Kv)$ exceeds the upper bound for $\vartheta(v)$. Hence

$$q \le K(N_t + H) < X$$

eventually. If $G(X)$ denotes the largest gap between consecutive primes with both endpoints below X, the construction gives

$$G(X) > C \frac{\log X \, \log_2 X}{\log_4 X}$$

for every fixed $C > 0$ and all sufficiently large X.

The estimates have enough room to make the dependence on X stronger. The restriction $k \le x/\ell^2$ was chosen to make all losses tend to zero automatically. It is useful now to inspect the finite inequalities instead of treating that restriction as part of the mechanism. The detector has bounded size M, and for an absolute K the energy bound is

$$\mathscr{E}(F) \le KkZ.$$

The insertion ratios also give $Z \ge 2^{-k}$. These estimates remain available for the longer intervals considered below, where $H \le x^2$ and $k\ell/H \to 0$.

The uniformity of the one-target estimates follows directly from the same Euler integral. For $x \le H \le x^2$,

$$L \le \log H \le 2L, \qquad \log \frac{\log y}{\log H} \ge L - 3\ell - O(1).$$

Thus c_* stays bounded above and below by positive absolute constants, and M and the energy constant can be fixed independently of the final interval length in this range. The control of used-source mass depends on $k\ell/H$, which still tends to zero for the new choice of H. The fourth-moment sieve-error estimate was already small uniformly for all $k \le x$.

Keep

$$y = e^{x/L^2}, \qquad B = x/100, \qquad D = e^B, \qquad T = \lfloor e^x \rfloor.$$

For sufficiently large x, the weighted sieve error is at most $Z/16$. Using $J_1 \le 2x/L^2$ and $\gamma \le 2$, the preceding estimates yield the explicit margin

$$\mathbb{E}R^2 - \mathbb{E}\left(R^2 \sum_i S_i^2\right) \ge \left(1 - 200K\frac{k}{x} - \frac{80000K}{L^2}\frac{k}{x} - \frac{1}{8}\right) Z.$$

The first loss is the norm removed by the filter, the second comes from the unhit-boundary estimate, and the last comes from replacing unhit indicators by finite sieve amplitudes. Only the first requires the ratio k/x to be a sufficiently small constant. The other two are controlled by the large source-cutoff ratio and the large value of L.

Choose a small absolute $\delta > 0$ with

$$200K\delta \le 1/8.$$

If $k \le \delta x$, the margin is at least $Z/2$ for all sufficiently large x. There is no need for k/x itself to tend to zero. A fixed fraction of the available spectral budget can be spent on the detector product.

The arithmetic errors allow the same enlargement. To avoid tracking the vanishing exponents in the coefficient estimates, bound their combined error Δ a little more crudely. For large x, the previous coefficient bounds imply

$$L_R \le M^k D^3, \qquad L_S \le D^2.$$

Since $T \ge e^x/2$,

$$\Delta \le 2(1+k)M^{2k}e^{-9x/10}.$$

Dividing by $Z \ge 2^{-k}$ gives

$$\frac{\Delta}{Z} \le 2(1+k)e^{-9x/10}(2M^2)^k.$$

Choose δ additionally so that

$$\delta \log(2M^2) \le 1/10.$$

Then, for $k \le \delta x$,

$$\frac{\Delta}{Z} \le 2(1+x)e^{-4x/5} \longrightarrow 0.$$

Thus the positive model margin still transfers to actual integers. The argument can handle a small absolute multiple of x remaining targets. This is a statement about the finite estimates used in the construction; it does not rely on substituting a growing constant into the fixed-C conclusion.

The preliminary sieve can now be retuned to use this larger survivor allowance. Take

$$H = \left\lfloor \kappa \frac{xL^2 h}{\ell^2} \right\rfloor,$$

where $\kappa > 0$ is a sufficiently small absolute constant. Use

$$w = \lfloor L^4 \rfloor, \qquad u_0 = \frac{5\ell}{h}, \qquad z = \left\lfloor e^{L/u_0} \right\rfloor.$$

The smoothness parameter has moved from $5h/\log h$ to $5\ell/h$. This makes the smooth survivors much rarer, at the cost of a larger surviving fraction among the prime candidates. The available linear survivor budget is what permits that tradeoff.

For large x,

$$w < z < \lfloor x \rfloor, \qquad H \le \lfloor x \rfloor w, \qquad x \le H \le x^2.$$

Also,

$$\frac{x\ell}{H} \asymp \frac{\ell^3}{\kappa L^2 h} \longrightarrow 0.$$

Hence every $k \le \delta x$ obeys the used-source condition needed above. The larger value of H causes no difficulty for the detector or the categorical model: all source primes still exceed every offset difference, and y remains much larger than H^2.

Choose zero residues for primes at most w and for primes between z and $\lfloor x \rfloor$. For the intermediate primes, it is simplest to choose uniformly among all residue classes. Every integer then survives the random part with exactly the same probability

$$\sigma = \prod_{w<p\le z} (1 - 1/p),$$

so no correction factor depending on its prime divisors is needed.

The classification of surviving composites is unchanged. A prime factor in $(z, \lfloor x \rfloor]$ is excluded by the zero sieve. A composite with a prime factor above $\lfloor x \rfloor$ has a complementary factor at most $H/\lfloor x \rfloor \le w$, which is excluded by the small-prime zero sieve. Every surviving composite is therefore z-smooth. The count

$$k \le \sigma\big(\pi(H) + \Psi(H, z)\big)$$

is valid for some deterministic choice of the intermediate residues, and includes prime powers and the integer one among the smooth candidates.

Mertens' estimate gives

$$\sigma \ll \frac{\log w}{\log z} \ll \frac{u_0 \ell}{L} \ll \frac{\ell^2}{Lh}.$$

For the smooth-number estimate, put $v = \log H / \log z$. Then $v \geq u_0 = 5\ell/h$, while $v \leq 4u_0$ for sufficiently large x. The same Rankin argument applies, and

$$\Psi(H, z) \leq 4H \exp(-v \log v + O(v)) \leq He^{-4\ell} = H/L^4.$$

Indeed, $u_0 \log u_0 = (5 + o(1))\ell$, and the $O(u_0)$ loss is $o(\ell)$. Increasing v only improves the exponent once v is sufficiently large. The floor functions change none of these estimates.

Since $\log H \asymp L$, it follows that

$$\begin{aligned} k &\ll \frac{\ell^2}{Lh}\left(\frac{H}{L} + \frac{H}{L^4}\right) \\ &\ll \kappa x. \end{aligned}$$

All constants here are absolute. Choosing κ small enough makes $k \leq \delta x$, which is exactly the enlarged range supported by the detector, filter, and arithmetic transfer. Both adjustments are needed: enlarging the survivor budget alone does not automatically give the longer interval, and the more restrictive smoothness cutoff must be paired with that budget.

The same construction now produces a composite interval of length

$$H = \left\lfloor \kappa \frac{xL^2 h}{\ell^2} \right\rfloor$$

with starting point at most $2e^{5x}$. Taking $x = (\log X)/6$ and using the same endpoint argument gives

$$H \sim \frac{\kappa}{6} \frac{\log X \, (\log_2 X)^2 \, \log_4 X}{(\log_3 X)^2}.$$

After absorbing the floor and the fixed comparison constants, the resulting bound is

$$G(X) > c \, \frac{\log X \, (\log_2 X)^2 \, \log_4 X}{(\log_3 X)^2}$$

for some absolute $c > 0$ and all sufficiently large X. Relative to the initial target, the permitted multiplier is

$$C(X) = c \, \frac{\log_2 X \, (\log_4 X)^2}{(\log_3 X)^2},$$

which tends to infinity.

The construction locates a complete cover without explicitly selecting all its divisors in advance. The preliminary sieve leaves a manageable set of offsets; the harmonic detector makes their simultaneous hit condition inexpensive in source energy; the gradual source cutoff preserves enough of that condition in a short arithmetic expansion; and the one-target fourth-moment comparison makes the finite prime score usable under the resulting global weight. The height is governed by $Q_0 e^x$, while the much larger source period serves only to organize the proof of the weighted inequality.

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